

# Inverse & Composite Functions



## Composing functions

- 1 Given  $f(x) = 2x + 3$  and  $g(x) = x^2 - 1$ , find  $(f \circ g)(x)$  and  $(g \circ f)(x)$ .
  - 2 Given  $f(x) = \sqrt{x + 4}$  and  $g(x) = x^2 - 4$ , find  $(f \circ g)(2)$  and state the domain restriction that makes  $f \circ g$  defined.
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## Decomposing a composite function

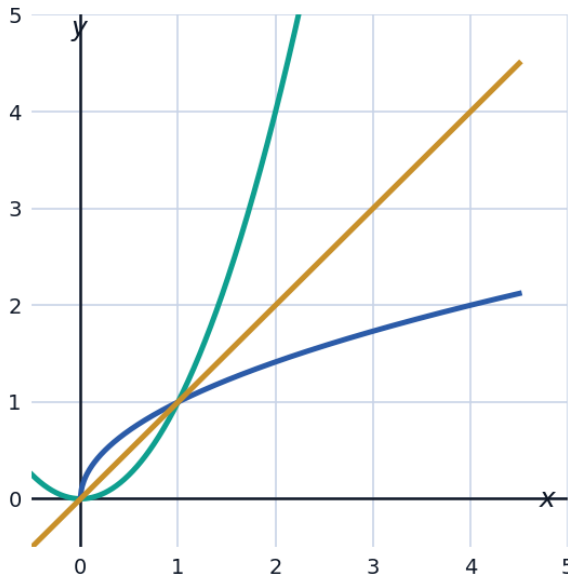
- 3 Write  $h(x) = (3x - 5)^4$  as a composition  $h(x) = f(g(x))$  for two simpler functions  $f$  and  $g$ .
  - 4 Write  $h(x) = \sqrt{x^2 + 1}$  as a composition of two functions.
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## Verifying inverse functions

- 5 Verify algebraically that  $f(x) = \frac{x-2}{3}$  and  $g(x) = 3x + 2$  are inverses of each other.
  - 6 Find the inverse of  $f(x) = 2x^3 + 1$ , and verify that  $f(f^{-1}(x)) = x$  for one value, say  $x = 17$ .
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## Visualizing inverse functions as reflections

- 7 The graph shows  $f(x) = x^2$  (restricted to  $x \geq 0$ ) and its inverse  $f^{-1}(x) = \sqrt{x}$ , which are reflections of each other across the line  $y = x$ .



- a State the domain restriction placed on  $f(x) = x^2$  needed for it to have an inverse, and explain why.
- b Confirm from the graph that  $f$  and  $f^{-1}$  intersect on the line  $y = x$ , and find that intersection point algebraically.

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### Domain and range of composite and inverse functions

- 8 Given  $f(x) = \frac{1}{x-3}$  and  $g(x) = x + 1$ , find the domain of  $(f \circ g)(x)$ .
- 9 If  $f$  has domain  $[1, \infty)$  and range  $[0, \infty)$ , state the domain and range of  $f^{-1}$ .

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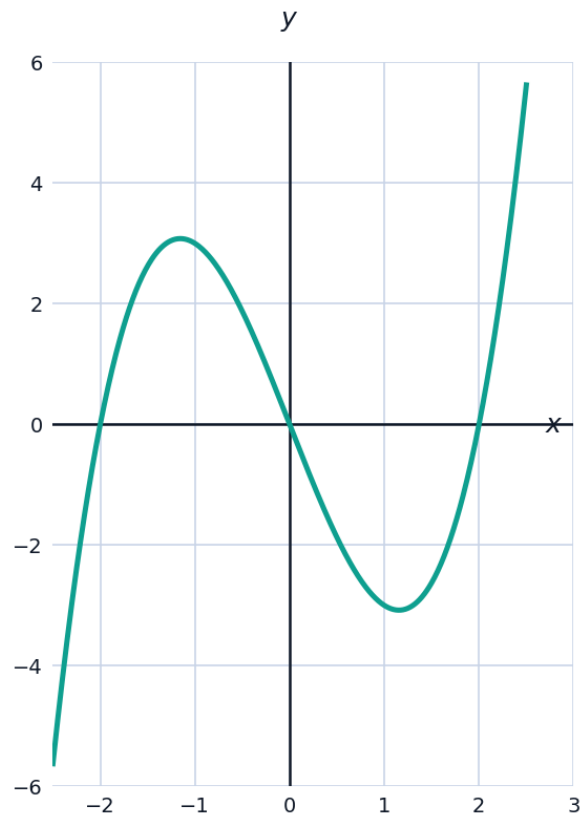
### Applying composite and inverse functions in context

- 10 A temperature-conversion function  $C(F) = \frac{5}{9}(F - 32)$  converts Fahrenheit to Celsius. Find  $C^{-1}(F)$  and interpret what it computes.

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### Using the horizontal line test

- 11 Determine whether  $f(x) = x^3 - 4x$  has an inverse function over its entire domain. Justify using the horizontal line test.



- 12 Explain why  $f(x) = 5x - 2$  passes the horizontal line test for all real  $x$ , and find its inverse.

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### Composite functions with restricted domains

- 13 Given  $f(x) = \frac{1}{x}$  and  $g(x) = x - 4$ , find  $(f \circ g)(x)$  and state its domain.